

1 Inferring causal networks for genomic trios with
2 high-dimensional confounding variables

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12 April 19, 2026

13 Abstract

14 Network inference under the principle of Mendelian randomization (PMR) enables the study
15 of causal relationships between two molecular phenotypes (e.g., gene expression) by leverag-
16 ing genetic variants as instrumental variables and forming a trio. In practice, inference is often
17 affected by numerous confounding variables, including other genes and demographic factors,
18 making it essential to robustly account for confounders. However, existing methods have lim-
19 ited ability to accommodate high-dimensional confounding. Here, we introduce Mendelian
20 Randomization for Genomic Networks (MRGN), a regression-based method under the PMR
21 for inferring causal relationships in genomic trios from individual-level data while accounting
22 for many confounding variables. MRGN extends beyond the canonical mediation model to
23 distinguish among multiple causal relationships. Through extensive simulations, we demon-
24 strate improved inference accuracy, robustness, and computational efficiency over existing ap-
25 proaches. We further apply MRGN to data from the Genotype-Tissue Expression (GTEx)
26 Consortium to investigate how genetic variants may regulate gene expression in cis and trans.

27 1 Introduction

28 An important goal in biology is to understand how genes regulate one another and how genetic
29 variation influences gene expression [1, 2, 3, 4]. Gene regulation involves complex networks of
30 high dimensions, which may be represented by directed networks where directed edges repre-
31 sent causal (or regulatory) relationships. Although the ultimate goal in biology is to reconstruct
32 gene regulatory networks at the genome level, it is often of interest to understand the causal re-
33 lationship between a molecular phenotype (e.g. the expression level) of two genes, or between
34 two molecular phenotypes (e.g., between gene expression and DNA methylation), which may
35 be influenced by many other variables such as other genes, other biological processes, demo-
36 graphic variables, etc.

37 To learn the (directed) causal relationship between two phenotypes, we can apply the prin-
 38 ciple of Mendelian randomization (PMR) and leverage the information in genetic variants. The
 39 PMR assumes that genotypes in a natural population are randomly assigned, thus observational
 40 omics data can be viewed as being collected from a randomized experiment [5, 6]. This implies
 41 that an edge connecting a genotype to a phenotype will always point to the phenotype, but not
 42 the other way around [7, 8]. As a result, this constraint on edge direction makes it possible to
 43 break the symmetry of association and draw inferences about the regulatory structure among
 44 phenotypes [5, 7, 9].

45 Methods based on the PMR framework differ fundamentally from classical Mendelian ran-
 46 domization (MR) methods. Traditional MR focuses on estimating causal effects and depends
 47 on three key assumptions [5, 10, 11]: (i) relevance, where the genetic variant is associated with
 48 the molecular phenotype; (ii) independence, where the variant is independent of unmeasured
 49 confounders due to random allele segregation; and (iii) exclusion restriction, where the variant
 50 influences downstream phenotypes only through the intermediate phenotype, and not through
 51 other pathways. In contrast, our PMR-based framework does not require these assumptions to
 52 hold in full, nor is causal effect estimation its primary objective. Instead, following our earlier
 53 work [7, 12, 13], we use the PMR as a structural constraint that limits edge directions in the
 54 network. Specifically, edges from molecular phenotypes to genetic variants are not allowed in
 55 the network.

56 Many methods have been developed under assumptions similar to the PMR [14, 15, 16,
 57 17, 7, 18, 19, 20], although some of these methods are interested mostly in the mediation
 58 model $V \rightarrow T_1 \rightarrow T_2$, where V is the genetic variant, T_1 and T_2 are the variables of interest.
 59 In this model, the mediator T_1 relays the regulatory information from V to the outcome T_2 ;
 60 for example, a genetic variant regulates the expression level of a transcription factor, and its
 61 reduced expression leads to its downstream target gene being under-expressed. However, other
 62 causal relationships may exist among the three variables.

63 We previously developed MRPC, a method that goes beyond mediation and infers five
64 mutually exclusive, basic causal models for a trio under the PMR (Figure 1a) [7, 12, 13]. They
65 include a null model ($V \rightarrow T_1$ and no relationship between T_1 and T_2 , denoted M0) and four
66 non-null models: mediation (denoted M1), v-structure (or reverse causation; $V \rightarrow T_1 \leftarrow T_2$,
67 denoted M2), conditional independence (or horizontal pleiotropy; $T_1 \leftarrow V \rightarrow T_2$, denoted
68 M3), and a fully connected model ($T_1 \leftarrow V \rightarrow T_2$ and $T_1 \leftrightarrow T_2$, denoted M4). The true model
69 behind M4 may be directed: the edge between T_1 and T_2 is $T_1 \rightarrow T_2$ or $T_1 \leftarrow T_2$, but these
70 two directions result in the same likelihood for the trio and are therefore Markov equivalent.
71 Without additional information, they are indistinguishable and only a bidirected/undirected
72 edge can be inferred for M4.

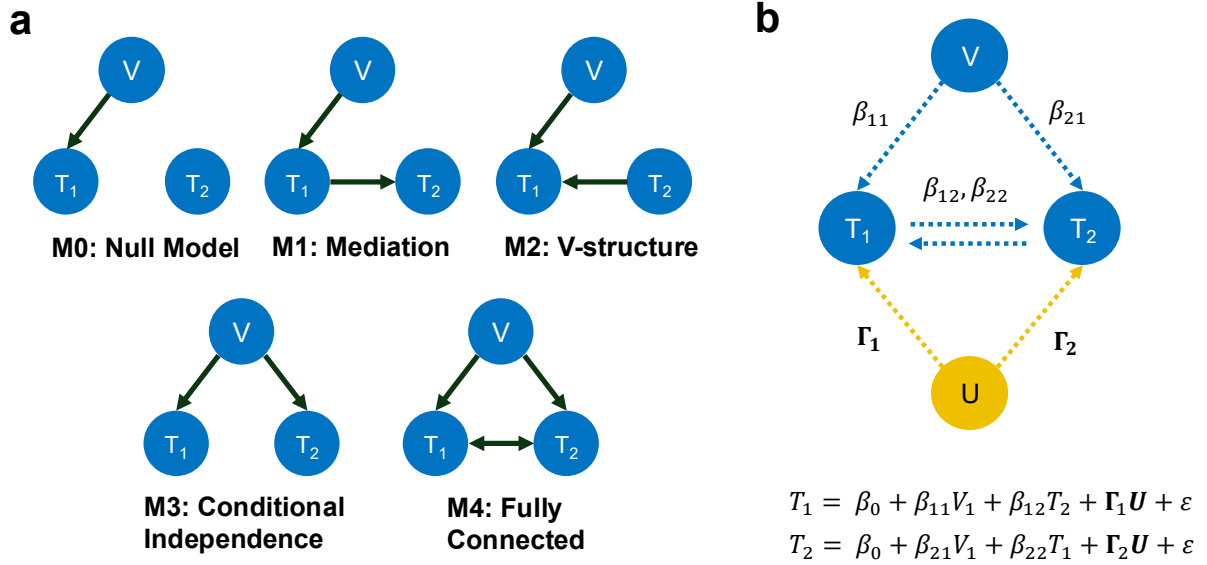


Figure 1: Causal models for a genetic trio. (a) The five basic causal models for a trio that may be inferred under the principle of Mendelian randomization (adapted from [7]). (b) A genomic trio includes a genetic variant (V), two molecular phenotypes (T_1 and T_2), and a set of confounders (U).

73 The PMR and the five basic causal models require that no confounding variables can ex-
74 plain away the directed edge between T_1 and T_2 , since this edge is our main interest. How-
75 ever, as mentioned earlier, many variables can influence the inference due to the complexity

76 of gene regulation. A confounding variable may influence the inference in different ways.
 77 Yang et al. [16] proposed three types of confounding C for the $T_1 - T_2$ edge: as a con-
 78 founder ($T_1 \leftarrow C \rightarrow T_2$), an intermediate variable ($T_1 \rightarrow C \rightarrow T_2$), or a common child
 79 ($T_1 \rightarrow C \leftarrow T_2$). Specifically, a confounder is a common parent to both T_1 and T_2 and can
 80 induce a false edge between T_1 and T_2 when unaccounted for. This is similar to T_1 and T_2
 81 being marginally dependent in M3. An intermediate variable acts like a mediator between T_1
 82 and T_2 and can potentially explain away a true $T_1 \rightarrow T_2$ edge when this intermediate variable is
 83 accounted for in the inference, leading to a false negative conclusion. By contrast, a common
 84 child can induce a false edge between T_1 and T_2 when accounted for, leading to a false positive
 85 conclusion. This is because when the common child is accounted for in the inference, T_1 and
 86 T_2 become conditionally dependent, similar to their relationship in M2.

87 Accounting for confounding variables is therefore crucial in causal network inference [21,
 88 16, 22, 23, 24]. Two basic approaches have emerged to address latent confounding between an
 89 exposure and outcome. In the first approach, candidate confounding variables are incorporated
 90 into the graph as additional nodes in the network and also treated in the same way as other
 91 nodes; our MRPC method and several other methods take this approach [7, 23]. However, this
 92 approach is computationally intensive, as the computational complexity for graph inference
 93 grows super exponentially with the number of nodes. The second approach focuses on identi-
 94 fying and accounting for confounders [21, 16, 22]. This approach includes all the confounders
 95 as a whole in the tests for edge presence, and does not increase the size of the network. For
 96 example, we can include the confounders when testing the $T_1 - T_2$ edge in a trio. The trio re-
 97 mains a network of three nodes, but the tests are conditioned on these confounders. However,
 98 these methods focus on the mediation model.

99 Herein we present Mendelian Randomization for Genomic Networks (MRGN), a novel
 100 machine learning method for inferring the causal model of a trio from individual-level data in
 101 the presence of potentially hundreds of confounding variables. MRGN builds on MRPC and

102 infers the five basic causal models. We take the second approach to confounding described
 103 above and focus on accounting for confounders, and not other types of confounding variables.
 104 This approach allows for the inclusion of many confounding variables all at once, while main-
 105 taining high power and fast computation times. We demonstrate the performance of MRGN
 106 and compare to other methods through extensive simulation studies. We further assess this
 107 method in our study of the cis- and trans-regulatory modes of genetic variants, using the whole
 108 blood data from the Genotype-Tissue Expression (GTEx) Consortium [25].

109 **2 Methods**

110 **2.1 The MRGN model**

111 The five causal models for a trio can each be described by a set of marginal and conditional
 112 dependence relations (Figure 1b; Table 1), which may be encoded in the coefficients of two
 113 regression models that account for confounders, assuming that T_1 and T_2 are centered so that
 114 the intercept is zero:

$$T_1 = \beta_{11}V + \beta_{12}T_2 + \mathbf{\Gamma}_1\mathbf{U} + \epsilon_1; \quad (1)$$

$$T_2 = \beta_{21}V + \beta_{22}T_1 + \mathbf{\Gamma}_2\mathbf{U} + \epsilon_2, \quad (2)$$

115 where β s are coefficients of interest, vector $\mathbf{\Gamma}_i$ the coefficients for the confounder vector $\mathbf{U}_i$,
 116 and ϵ_i the random error that follows a normal distribution with mean 0 and standard deviation
 117 σ_i , where $i = 1, 2$.

118 Our conditional independence tests test whether each β is zero, as each β represents the
 119 association given the presence of other variables. For example, in the mediation model of
 120 $V \rightarrow T_1 \rightarrow T_2$, any two variables are dependent on each other, but T_2 and V are conditionally
 121 independent given T_1 . These relations translate to (i) β_{21} being zero, which represents the

122 association between V and T_2 conditioned on T_1 and the confounders U , and (ii) the other
123 three β s being nonzero, which represent those associations not being explained away by other
124 variables. Similarly, models M0 and M3 are uniquely determined by conditional associations,
125 for which we only need to conduct conditional independence tests.

126 In the v-structure model M2, consider the configuration $V \rightarrow T_1 \leftarrow T_2$. As the parents of
127 T_1 , nodes V and T_2 are marginally independent but conditionally dependent given the presence
128 of their child T_1 . Therefore, all the β s, which represent conditional associations, should be
129 nonzero. Similarly, all the β s are also nonzero in the fully connected model M4. The distinction
130 between M2 and M4 lies in the marginal associations: V and T_2 are marginally independent
131 in M2 but are marginally dependent in M4. MRGN therefore performs additional tests on
132 marginal dependence to distinguish these two models.

133 MRGN performs the standard t tests on the regression coefficients for conditional depen-
134 dence and the standard test on Pearson correlation for marginal dependence. Since the number
135 of tests for the edges of interest is at most six, we set the significance level for each test to
136 be 0.01, which is equivalent to Bonferroni correction at a familywise type I error rate of 0.05.
137 When the data is not informative enough for any of the basic models to be inferred, MRGN
138 categorizes the trio as “other”.

Table 1: Conditional and marginal independence tests in MRGN. Regression coefficients β_{11} and β_{12} (from Equation 1) and β_{21} and β_{22} (from Equation 2) are used to test conditional independence. Marginal independence can be tested through correlations r_{V,T_1} and r_{V,T_2} . The first three models each have two possible configurations.

		Conditional Test				Marginal Test	
Null Hypothesis		$T_1 \perp\!\!\!\perp V$	$T_1 \perp\!\!\!\perp T_2$	$T_2 \perp\!\!\!\perp V$	$T_2 \perp\!\!\!\perp T_1$	$V \perp\!\!\!\perp T_1$	$V \perp\!\!\!\perp T_2$
Conditioned on		$T_2, \mathbf{U}$	$V, \mathbf{U}$	$T_1, \mathbf{U}$	$V, \mathbf{U}$	–	–
Parameter to Test		β_{11}	β_{12}	β_{21}	β_{22}	r_{V,T_1}	r_{V,T_2}
Model	Configuration						
M0	$V \rightarrow T_1; T_2$ is singleton	$\neq 0$	$= 0$	$= 0$	$= 0$	–	–
	$V \rightarrow T_2; T_1$ is singleton	$= 0$	$= 0$	$\neq 0$	$= 0$	–	–
M1	$V \rightarrow T_1 \rightarrow T_2$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$	–	–
	$V \rightarrow T_2 \rightarrow T_1$	$= 0$	$\neq 0$	$\neq 0$	$\neq 0$	–	–
M2	$V \rightarrow T_1 \leftarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$
	$V \rightarrow T_2 \leftarrow T_1$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$
M3	$T_1 \leftarrow V \rightarrow T_2$	$\neq 0$	$= 0$	$\neq 0$	$= 0$	–	–
M4	$T_1 \leftarrow V \rightarrow T_2;$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$
	$T_1 \longleftrightarrow T_2$						

2.2 Confounder selection

We perform confounder selection before causal network inference, using a two-step procedure similar to that in the GMAC method [16]. First, we identify candidate confounders from a pool of potential confounding variables. Confounders, which are defined as parents of T_1 and T_2 , should be independent of V . We therefore exclude variables that are marginally associated

144 with V . Next, we select confounders based on their statistical association with either T_1 or T_2 .
 145 Similar to [16], we test whether the regression of both T_1 and T_2 on a candidate confounder is
 146 statistically significant. Both steps allow for an additional step of the q -value method to control
 147 the false discovery rate (FDR) [26].

148 In our R implementation, we substantially improve computational efficiency in two key
 149 ways. First, we implemented parallelization in both steps, allowing for faster evaluation when
 150 the pool of confounders is large. Second, we used the R package *propagate* for correlation
 151 matrix computation by partitioning the problem into smaller, memory-efficient blocks [27],
 152 which scales more efficiently to large genomic datasets.

153 **2.3 Simulation study**

154 **Simulation setup**

155 Each simulated trio consists of a bi-allelic genetic variant V with three genotypes and two
 156 genes (T_1, T_2) connected under one of the five causal models. Nodes in a trio are generated in
 157 topological order. The genotype V_i is simulated first under Hardy–Weinberg equilibrium:

$$\Pr(V_i = 0) = (1 - \theta)^2, \quad \Pr(V_i = 1) = 2\theta(1 - \theta), \quad \Pr(V_i = 2) = \theta^2,$$

158 where the minor allele frequency θ is sampled uniformly from $(0.01, 0.5)$, such that roughly
 159 10% of trios contain a less common allele with a $< 5\%$ frequency.

160 Each remaining node Y_j (a gene or a confounder) is sampled from

$$Y_j \sim N\left(\sum_{i \in S_j} \beta_i Y_i, \sigma^2\right),$$

161 where S_j is the parent set of Y_j , which may include V , the other gene, or confounders. The
 162 effect size of V on its target genes is sampled uniformly from $[0.5, 1.5]$ [16]. The effect size

between T_1 and T_2 is sampled uniformly from $[0.5, 1]$. Residual variance is set to $\sigma = c\beta$, where β is the T_1 – T_2 effect and c sampled uniformly from $[0.3, 1.5]$.

For **M4**, the true edge between T_1 and T_2 can be in either direction. For simplicity, we fix all M4 trios to the configuration $T_1 \leftarrow V \rightarrow T_2$ with $T_1 \rightarrow T_2$.

For each trio, we further include three types of confounding variables: a varying number of confounders (between 1 and 50), one intermediate variable, and one common child. Effect sizes for these variables are sampled independently: confounder and intermediate effects from $[0.15, 0.5]$ and common-child effects from $[0.5, 1.5]$, each with a 50% probability of being negative.

We simulate 1,800 trios by sampling from the five causal models with equal probability, and obtain approximately 360 trios per model. For each trio, we draw a distinct combination of effect sizes, noise level, and minor allele frequency. The sample size is fixed at $n = 670$, matching the whole-blood sample size of GTEx v8 [25] in our real-data application.

Performance metrics

We evaluated three sets of recall (i.e., power) and precision to quantify the performance on the entire model of a trio, on individual edges and specifically on the $T_1 - T_2$ edge.

1. On the causal model type: a trio is considered correctly inferred when its inferred model type matches its true type, thus requiring all edges to be inferred correctly.

$$\text{Recall}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios of the causal model M}}; \quad (3)$$

$$\text{Precision}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios inferred to be of the causal model M}}. \quad (4)$$

2. On individual edges: we first calculated recall and precision based on individual edges in a single trio. Similar to our earlier work [7], we counted an edge with the wrong

183 direction as a half correct edge.

$$\text{Recall}_{\text{edges}} = \frac{\text{Number of correct edges in inferred graph}}{\text{Total edges in true graph}}; \quad (5)$$

$$\text{Precision}_{\text{edges}} = \frac{\text{Number of correct edges In inferred graph}}{\text{Total edges in inferred graph}}. \quad (6)$$

184 We then calculated the mean metric across all trios.

185 3. On the $T_1 - T_2$ edge (allowing for comparison with the GMAC method): the direct of
186 this edge was ignored for this evaluation. Across all trios,

$$\text{Recall}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with a true } T_1 - T_2 \text{ edge}}; \quad (7)$$

$$\text{Precision}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with an inferred } T_1 - T_2 \text{ edge}}. \quad (8)$$

187 3 Results

188 3.1 Simulation Results

189 MRGN improves power and precision to detect edges in the presence of confounding

190 Before applying MRGN, we examined two approaches to confounder selection: controlling the
191 FDR at 5% with the q -value method (denote “CS- q ”), and not correcting for multiple testing
192 but instead setting the type I error rate of each individual test at $\alpha < 0.01$ (denoted “CS- α ”).
193 We evaluated their ability to identify the true confounders, measured by precision and recall.
194 Here, recall is defined as the proportion of true confounders correctly identified and selected
195 for the inference model, and precision is the proportion of true confounders selected out of
196 all confounders selected. Overall, both methods achieved recall greater than 90% when the
197 number of true confounders per trio was modest (e.g., fewer than 25; see Figure 2). However,

198 CS-q showed reduced performance for trios with more true confounders, with average recall
 199 dropping to approximately 65%. Conversely, as expected, CS- α achieved nearly perfect re-
 200 call across all simulations, but at the expense of low precision. Whereas the actual FDR (i.e.,
 201 1-precision) under CS- α was above 70% and increased to above 90% as the number of true
 202 confounders decreased, the actual FDR under CS-q was generally around 20%, already much
 203 higher than the expected level, and increased to 50% as the number of true confounders de-
 204 creased. This is likely due to the dependency among the tests: this step performs many tests
 205 involving the same variable or highly correlated variables.

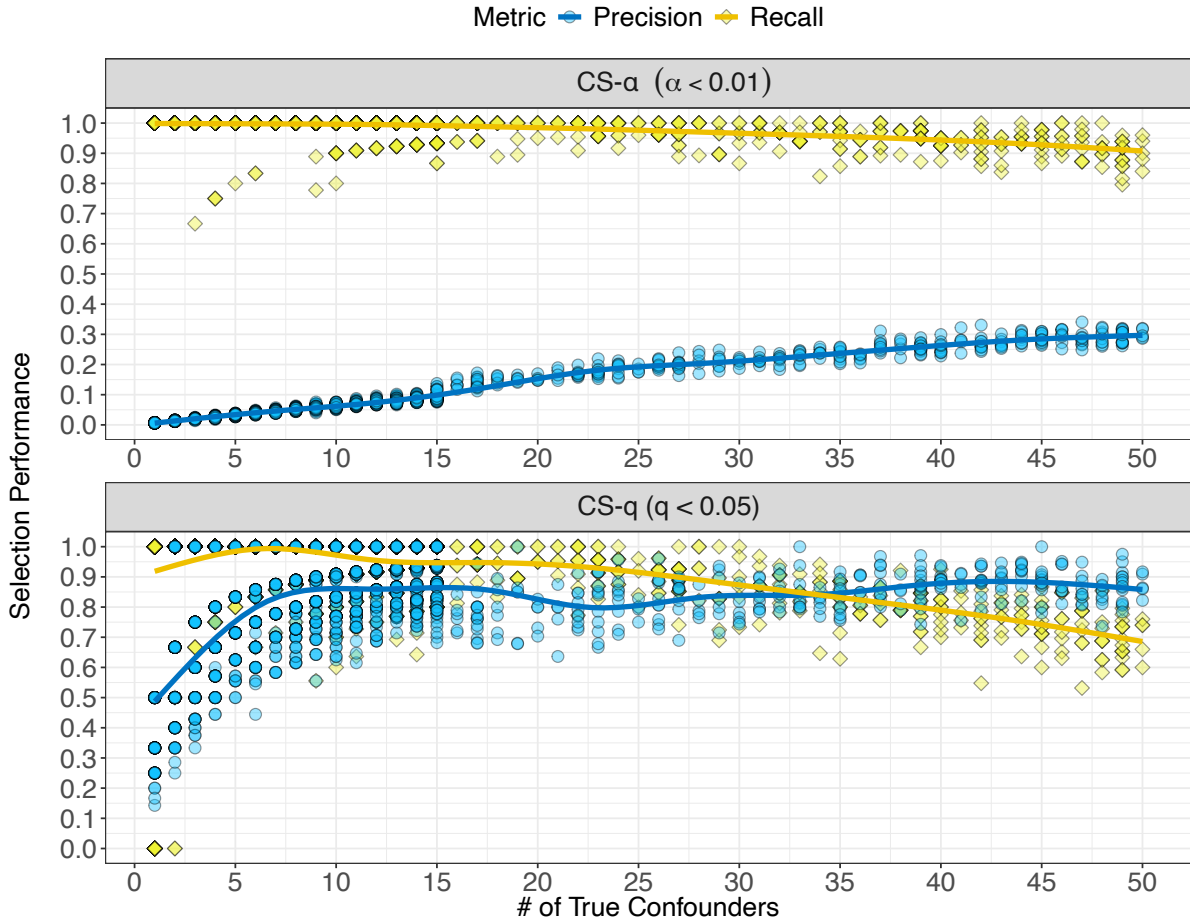


Figure 2: Precision and recall of confounder selection on simulated data by two multiple-testing correction methods: CS-q and CS- α . Each shape is a simulated data set.

206 We applied MRGN with these two methods for confounder selection. They led to qualita-

tively similar results in recall and precision on the inference of the trio (Table 2). Both versions achieved substantially higher recall for the mediation, v-structure, and fully connected models than MRPC did. All these models contain a $T_1 - T_2$ edge, indicating the sensitivity of MRGN in detecting the $T_1 - T_2$ edge. On the other hand, compared to MRPC, this sensitivity led to reduced power to detect the null and conditional independence models, which have no $T_1 - T_2$ edge. Remarkably, MRGN also achieved equivalent or better precision for nearly all models, as well as similar performance between the two confounder selection methods, demonstrating robustness to high false positive rates in confounders.

Table 2: Comparison of MRGN, MRPC and GMAC on simulated trios with confounders.

The oracle performance of MRGN using true confounders is shown to illustrate the upper bound due to noise in the data. For the other columns, the highest precision and recall in each row is highlighted. CS-q: confounder selection with the q -value method for FDR control at $q < 0.05$. CS- α : confounder selection with the type I error rate set at $\alpha < 0.01$ and no FDR control. MRPC cannot handle too many variables in reasonable runtime; we therefore did not run MRPC with the many confounders identified from CS- α . Rows 1-5 show model type-based performance, row 6 for all three edges in a trio, and the last row for only the $T_1 - T_2$ edge. * indicates that similar values are produced by the GMAC method.

	MRGN (True Confs.)		MRGN (CS-q)		MRGN (CS- α)		MRPC (CS-q)	
	Recall	Prec.	Recall	Prec.	Recall	Prec.	Recall	Prec.
Null Model (M0)	0.562	0.995	0.452	1.000	0.477	0.951	0.836	0.532
Mediation (M1)	0.982	0.516	0.877	0.438	0.883	0.429	0.097	0.147
V-structure (M2)	0.919	0.985	0.892	0.925	0.842	0.968	0.377	0.443
Cond. Indep. (M3)	0.515	0.989	0.283	0.910	0.305	0.872	0.609	0.828
Fully Connect. (M4)	0.600	0.566	0.501	0.425	0.464	0.419	0.217	0.592
All Edges	0.945	0.893	0.931	0.864	0.911	0.857	0.770	0.892
$T_1 - T_2$	1.000	0.767	0.994*	0.703*	0.9787	0.7071	0.7884	0.916

At the edge level, MRGN recovered more true edges than MRPC did, while maintaining a high degree of precision. MRGN and GMAC performed similarly well in recall and maintained

consistently high recall for the $T_1 - T_2$ edge, even under substantial confounding, with precision remaining relatively unchanged (Supplementary Figure 1). This is because the GMAC test for mediation is a special case of our method. The similar performance on the $T_1 - T_2$ edge further shows that the notion of “mediation” detected by GMAC is broad and an inferred mediation edge by GMAC may correspond to different underlying causal models. Additionally, recall and precision for all the edges in a trio remained high, indicating that MRGN can accurately infer most edges, except in cases with many (> 35) confounders, where recall declined but still exceeded that of MRPC (Supplementary Figure 2).

The impact of confounder selection on causal network inference

When we examined how confounder selection affected causal inference performance, we found no clear relationship between the number of falsely included confounders and MRGN’s ability to accurately infer causal edges (Figure 3a). In contrast, falsely omitted confounders had a strong negative impact on causal inference performance (Figure 3b).

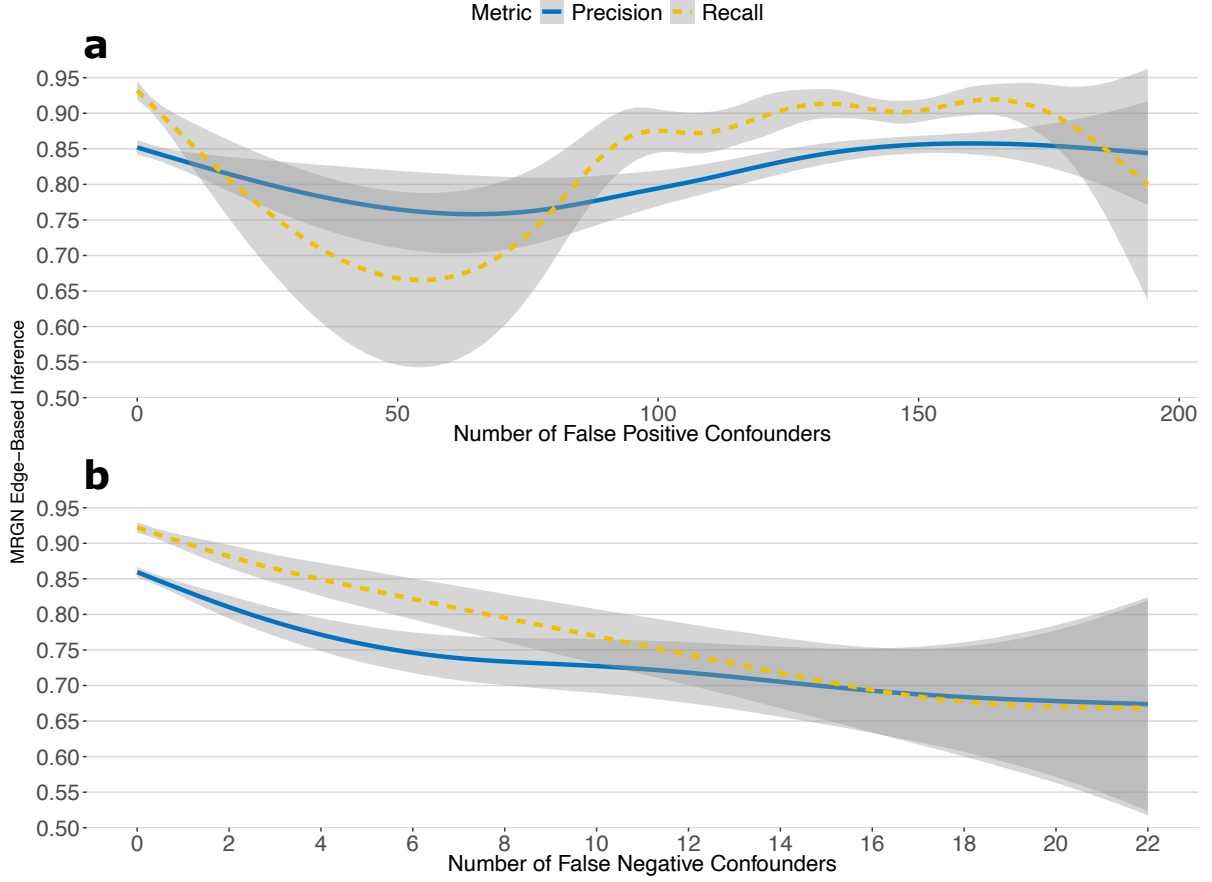


Figure 3: Relationship between confounder selection and causal inference performance measured as precision and recall of edge prediction. (a) Edge-based performance of MRGN on simulated trios as a function of the number of falsely included confounders. (b) Edge-based performance of MRGN on simulated trios as a function of the number of true confounders falsely omitted from the analysis.

We further observed substantially higher actual type I error rates for the $T_1 - T_2$ edge under the GMAC method than those reported by [16] (Supplementary Table 2). Their actual type I error rates were consistent with the significance level, such as $\alpha = 0.01$ or 0.05 . However, the actual type I error rates in our simulation using GMAC are 0.73 at $\alpha = 0.05$ and 0.63 at $\alpha = 0.01$. Using MRGN, our actual type I error rates are also around 0.6 (with $\alpha = 0.01$ in causal network inference). We attribute these differences to several distinctions between our simulation framework and theirs.

First, the number of confounding variables is much larger in our simulation than in theirs.

238 We simulated a substantially larger number of confounders (between 1 and 50) for each trio.
239 We further aggregated the confounding variables from all the trios for confounder selection,
240 resulting in a large candidate pool of more than 10,000 variables. In contrast, [16] simulated a
241 much smaller pool of 350 confounders, yielding an average of approximately two confounders
242 per trio, with many trios having no confounders. Second, we simulated two additional con-
243 founding variables for each trio: one common child and one intermediate variable. These
244 variables were also included in the candidate covariate pool. The simulations in [16], however,
245 treated each type of confounding variable separately.

246 To validate the error rates reported by [16], we used their simulation design to examine
247 the effect of confounders on type I error. We simulated 100 additional trios for each of seven
248 scenarios containing 2, 4, 6, 8, 10, 12, or 15 confounders, excluding common child and in-
249 termediate variables. We aggregated all the confounders in the candidate pool, and applied
250 GMAC with CS-q for confounder selection. The actual type I error rate under two true con-
251 founders is comparable to that reported by [16] (Supplementary Figure 4), whereas all other
252 scenarios showed that the actual type I error rate scaled with the number of confounders and the
253 size of the candidate pool. These results demonstrate that more confounding leads to markedly
254 inflated false positive $T_1 - T_2$ edges and decreased power to detect the null (M1) and conditional
255 independence (M3) models by both GMAC and MRGN.

256 Runtime comparison

257 Since our analysis involves two major steps: confounder selection and causal network infer-
258 ence, we report the computation time for each step separately. We conducted a comparative
259 analysis of the time required for confounder selection across varying sizes of the confounder
260 covariate pool (100, 500, or 1000 covariates) and different numbers of trios for which con-
261 founders needed to be identified (500 or 1000), resulting in a total of six replicates. Notably, in
262 the most intensive case 1000 covariates and 1000 trios, our method accomplished confounder

selection within 16 minutes (Figure 4). This represents a notable gain in computation efficiency compared to the implementation provided by GMAC, which took approximately 28 minutes. Although our confounder selection procedure is similar to that of [16], our implementation is more efficient due to the use of the R package propagate [27], which enables fast correlation calculations between genes in each trio and the candidate covariate pool, significantly reducing the time required for confounder selection.

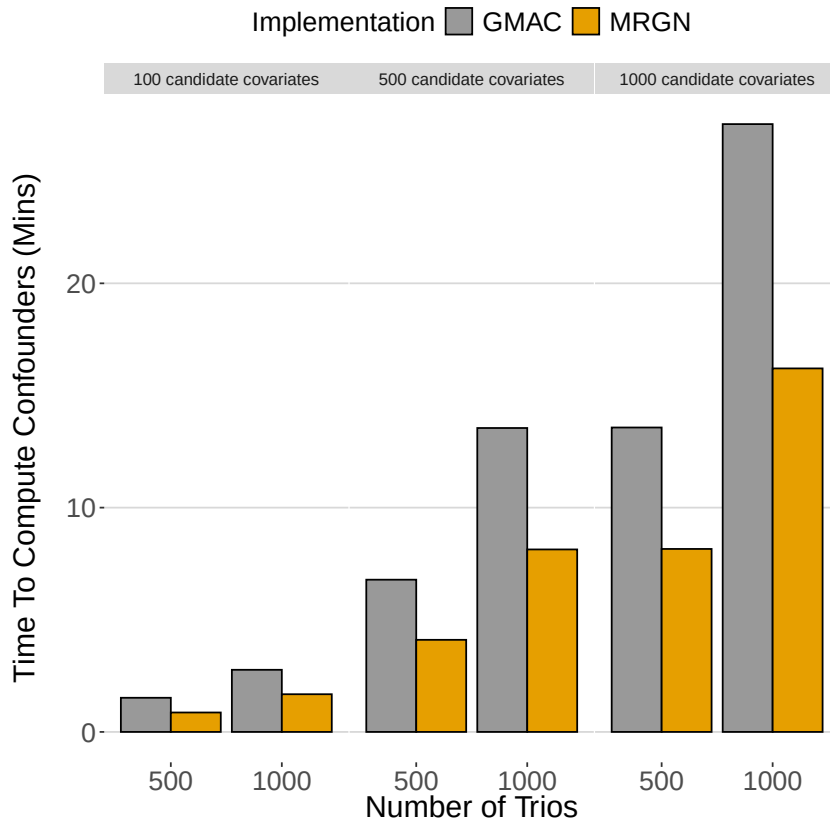


Figure 4: Comparison of computation times between our implementation of the CS-q method for confounder selection and that of the GMAC method [16]. Computation times are shown across different numbers of input trios and sizes of the candidate confounder pool.

For the step of causal network inference, we evaluated the runtime on all the 1,800 trios. MRGN is substantially faster than both MRPC and GMAC (Table 3). The median time to analyze a trio using the same set of confounders was 0.0065 seconds for MRGN, compared to 0.3557 seconds for MRPC and 0.7088 seconds for GMAC. This makes MRGN over 50 times

273 faster than MRPC and more than 100 times faster than GMAC, on average.

274 The extended inference time for MRPC is largely due to the ADDIS method used for FDR
275 control [13], which was particularly slow for some trios. When MRPC was applied using
276 the more conservative LOND method for FDR control [7], its runtime became comparable
277 to that of MRGN. For GMAC, we note that our comparison used the default single-threaded
278 implementation provided in its R package. Enabling the parallelization option available in
279 GMAC can substantially reduce its computational time.

Table 3: Inference times for MRGN, MRPC, and GMAC on simulated trios. We used the ADDIS method for FDR control in MRPC [13]. MRGN shows a substantially faster median inference time per trio compared to both MRPC-ADDIS and GMAC, demonstrating its computational efficiency in the presence of confounding.

Method	Median Time (s)	IQR (s)
MRGN	0.011	0.005
MRPC-ADDIS	0.460	9.012
GMAC	1.537	1.483

280 3.2 Application to GTEx data

281 The GTEx consortium has mapped a large number of genetic variants as expression quantitative
282 trait loci (eQTLs), which are associated the expression for most genes in the human genome
283 across many human tissue types [25]. Recent studies have focused on identifying cis-target
284 genes that mediate the relationship between the eQTL and its associated trans-target genes
285 [16]. We previously have applied the MRPC method to all the GTEx tissues to detect diverse
286 causal models for such trios: an eQTL and its associated cis- and trans-target genes, where the
287 genes are protein coding genes or long non-coding RNAs (lncRNAs) [13].

288 We implement the preprocessing and normalization pipeline from Section 2.2 to construct
289 trios for WholeBlood tissue from version 8 of GTEx. Each trio consists of an eQTL and its

highest association “cis” and “trans” target genes. We define cis genes as genes less than one megabase from the eQTL and trans genes as genes residing more than one megabase from the eQTL or on a different chromosome. As noted in Section 2.2 individual trios may share the same eQTL or the same gene, and the gene may be a cis-gene in one trio but a trans gene in another trio.

In GTEx version 8, the number of available samples for whole blood is 670. After preprocessing, 8,823 trios were identified. To eliminate bias induced by mismapping transcripts, we restricted our final set of trios to protein coding genes or long non-coding RNA genes [16]. This additional filtering step reduces the number of trios to 3,248. Many of eQTLs provided by the GTEx consortium contain missing values. Since the GMAC algorithm is unable to handle missing values, we used multiple correspondence analysis implemented in the *missMDA* package in *R* [28] to impute the missing genotypes of individuals using the collection of unique eQTLs from all GTEx trios analyzed for whole blood.

Confounding variable selection and filtering for GTEx trios

We performed principal component analysis (PCA) on the PEER-normalized, genome-wide gene expression data to derive 670 principal components (PCs) as candidate confounders. Each PC summarizes the regulatory effects of a large number of genes across the genome. Because each trio represents a small component of a larger regulatory network, PCs that correlate with a trio may capture potential confounding effects arising from pleiotropy. We applied MRGN using CS- α at a more relaxed criterion of $\alpha < 0.05$ than that used in the simulation, in order to include more potential confounders. For each trio, we identified a median of 51 confounders (SD = 6.38) (Figure 5a). We also included sex, platform, and PCR batch as additional, known confounders for each trio.

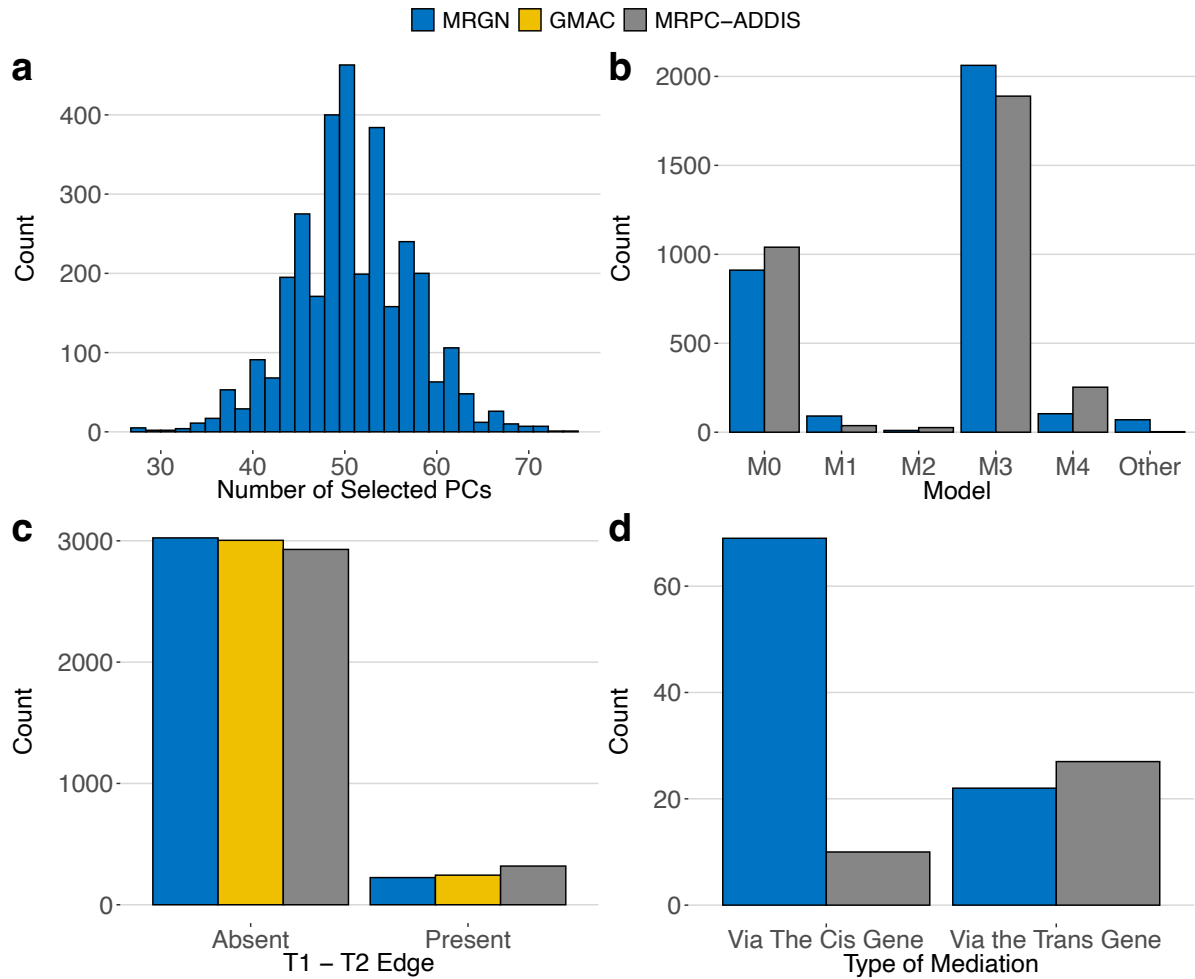


Figure 5: Comparative analysis of MRGN, MRPC [29] and GMAC [16] on GTEx human whole blood trios. We applied MRGN, GMAC, and MRPC (with ADDIS for FDR control) to genomic trios. (a) Histogram of the number of Principal Components (PCs) selected as confounders per trio. (b) Count of inferred causal models (M1–M4 and Other) comparing MRGN and MRPC. (c) Counts of trios with the $T_1 - T_2$ edge inferred as absent or present by the three methods. (d) Counts of two types of mediation models (via cis-gene vs. trans-gene) inferred by MRGN and GMAC.

314 MRGN identified 911 trios as the null model, 91 as mediation, 10 as v-structures, 2,062 as con-
 315 ditionally independent, 104 as fully connected, and 70 trios as “other” (Figure 5b), although the
 316 overall number of inferred $T_1 - T_2$ edges was similar to that from the other methods (Figure 5c).

317 Comparing with our previous analysis using MRPC [13], we found that MRGN identified more
 318 than twice as many cases of mediation and 173 additional cases of conditional independence.
 319 Similar to MRPC, MRGN also distinguishes two types of mediation (Figure 5d): (1) cis-gene
 320 mediation $V \rightarrow T_1 \rightarrow T_2$, where the cis-gene mediates the signal from genetic variant to the
 321 trans-gene, and (2) trans-gene mediation $V \rightarrow T_2 \rightarrow T_1$, where the trans-gene acts as the me-
 322 diator between the genetic variant and the cis-gene. Among the 91 trios identified as mediation
 323 by MRGN, 69 (75.5%) had the cis-gene as the mediator and 22 (24.5%) had the trans-gene as
 324 the mediator (Supplementary Table 4). Compared to MRPC, MRGN identified not only more
 325 mediation trios overall, but also a higher proportion of cis-gene mediation.

326 However, only 8 cis-gene mediation trios and 6 trans-gene mediation trios were identified
 327 by both methods. We investigated trios where MRGN inferred mediation but MRPC did not
 328 and found that these discrepancies were primarily driven by differences in the sets of selected
 329 confounders (see Examples 1 and 2 in Supplementary Text 1). In two illustrative cases, MRGN
 330 selected 40 and 43 confounders, whereas MRPC selected only three and none. When we
 331 reran MRGN using the same confounder set and inference model as MRPC, both methods
 332 produced identical results, confirming that accurate identification of confounders is essential
 333 for consistent causal inference.

334 **MRGN is robust to model misspecification for inference of mediation**

335 Based on our simulation and application findings, we conducted a secondary analysis to in-
 336 vestigate how model misspecification affects the inference of the mediation model, since this
 337 causal relationship has been the focus of many studies. In particular, we examined how inac-
 338 curacies in the inference model, particularly in terms of confounding variables, influence the
 339 power to detect mediation on the $T_1 - T_2$ edge. We base our simulations on 192 Whole Blood
 340 trios that were positive for mediation under the GMAC mediation test. We consider four sim-
 341 ulation scenarios: (i) the inference model included true confounders and a few false positive

342 confounders; (ii) the inference model included most true confounders, but missed a few true
343 confounders with weak effects; (iii) the inference model included only the true confounders;
344 and (iv) the inference model included only a few true confounders with strong effects, and
345 missed many true confounders with moderate to strong effects (See Supplementary Text 1 for
346 details and methods). This investigation is similar to the simulation study that examined the
347 impact of confounder selection on model inference, but further takes into consideration the
348 effect sizes of confounders here.

349 Recall is around 93% in each of the first three scenarios, indicating that when the inference
350 model contains most of the influential confounding variables, the correct inference for the
351 $T_1 - T_2$ edge can typically be recovered. Indeed, the first two scenarios represent relatively
352 minor changes in the confounding variables used in the inference model, either by omitting
353 or including weak confounders. On the other hand, even the correct model failed on a small
354 subset of trios due to noise in the data. In the final scenario, however, the inference model
355 omitted an average of 28 confounding variables ($SD \approx 7$), many of which had moderate to
356 strong confounding effects. Recall is reduced to as low as 70%, highlighting the critical need
357 to retain high-impact confounders for accurate causal network inference.

358 4 Discussion

359 In summary, we have developed MRGN to improve the power and accuracy of detecting key
360 causal relationships in diverse genomic trios, particularly in the presence of multiple confound-
361 ing variables. Our method adopts an extended interpretation of the PMR [7] and leverages
362 conditional and marginal hypothesis testing to identify five basic causal trio structures. A ma-
363 jor advantage of our approach is the reduction of the multiple testing burden; by adjusting for
364 selected confounders within a regression framework, MRGN can incorporate a large number
365 of confounders without the need to infer a graph larger than three nodes.

366 Our simulation results further underscore the critical importance of confounder selection in

causal inference. Omitting key confounders drastically reduces the power to detect mediation, leading to Type II errors and the loss of regulatory insights. Conversely, the inclusion of false-positive confounders has a negligible impact, particularly when their effects are weak. This suggests that in practical applications, a liberal strategy favoring the inclusion of many potential confounders is preferable. Ultimately, MRGN balances this requirement without excessive computational cost, making it a robust tool for causal network inference. However, when the sample size is not much larger than the number of identified confounders, including many false confounders can substantially increase the number of regression coefficients, leading to unreliable estimation. We are currently developing extensions of MRGN to address issues with a small sample size.

Constraint-based methods such as MRPC initialize learning by filtering edges from a fully connected graph structure. After removing edges with weak marginal relationships, the remaining edges are tested sequentially by conditioning on an increasing neighborhood of nodes [7, 30]. Critically, MRPC treats confounding variables as additional nodes within the graph. When confounders are included, the number of conditional tests increases significantly, and only edges with strong supporting evidence survive. Thus, MRPC inherently infers a larger network than MRGN for the same biological question. Additionally, the adaptive FDR can result in significance thresholds that deviate substantially from the fixed $\alpha = 0.01$ employed by MRGN. As demonstrated by our simulations, this causes MRPC to act more conservatively in the presence of confounding variables.

While our simulations and real data analysis demonstrate the flexibility of MRGN in robustly estimating causal relationships, the method is currently limited to trios containing a genetic variant that acts as an instrumental variable. Similar to MRPC, MRGN relaxes strict instrumental variable assumptions by empirically validating the association between the variant and the phenotype(s). We are currently extending this approach to develop a general MRGN algorithm capable of accommodating larger networks and addressing other classes of con-

393 founding variables.

394 **Data access**

395 Data from The Genotype-Tissue Expression (GTEx) Project version 8 were obtained from db-
396 GaP (<http://www.ncbi.nlm.nih.gov/gap>) through accession number phs000424.v8.pht002741.v8.
397 Open-source R packages used in this manuscript include MRGN (version 1.3.7.3); [https://](https://github.com/audreyfulab/MRGN)
398 github.com/audreyfulab/MRGN, MRPC (version 3.1.0); [https://cran.r-project.](https://cran.r-project.org/web/packages/MRPC/index.html)
399 [org/web/packages/MRPC/index.html](https://cran.r-project.org/web/packages/MRPC/index.html), and GMAC (version 3.1; [https://cran.](https://cran.r-project.org/web/packages/GMAC/index.html)
400 [r-project.org/web/packages/GMAC/index.html](https://cran.r-project.org/web/packages/GMAC/index.html). R code for simulations and
401 applications is available at [https://github.com/audreyfulab/MRGN_Manuscript_](https://github.com/audreyfulab/MRGN_Manuscript_Repo)
402 Repo. Detailed results generated by our analyses are included in the supplementary material.

403 **Competing interest statement**

404 The authors declare no competing interests.

405 **Acknowledgments**

406 We acknowledge the extensive computing support provided by the Research Computing and
407 Data Services from the Institute for Interdisciplinary Data Sciences at the University of Idaho.

408 This research is supported by the National Institutes of Health (NIGMS P20GM104420).

409 The Genotype-Tissue Expression (GTEx) Project was supported by the Common Fund of
410 the Office of the Director of the National Institutes of Health. Additional funds were provided
411 by the NCI, NHGRI, NHLBI, NIDA, NIMH, and NINDS. Donors were enrolled at Biospec-
412 imen Source Sites funded by NCI/Leidos Biomedical Research, Inc. subcontracts to the Na-
413 tional Disease Research Interchange (10XS170), Roswell Park Cancer Institute (10XS171),
414 and Science Care, Inc. (X10S172). The Laboratory, Data Analysis, and Coordinating Center
415 (LDACC) was funded through a contract (HHSN268201000029C) to the The Broad Institute,
416 Inc. Biorepository operations were funded through a Leidos Biomedical Research, Inc. sub-
417 contract to Van Andel Research Institute (10ST1035). Additional data repository and project
418 management were provided by Leidos Biomedical Research, Inc. (HHSN261200800001E).
419 The Brain Bank was supported supplements to University of Miami grant DA006227. Sta-
420 tistical Methods development grants were made to the University of Geneva (MH090941 &
421 MH101814), the University of Chicago (MH090951, MH090937, MH101825, & MH101820),
422 the University of North Carolina - Chapel Hill (MH090936), North Carolina State University
423 (MH101819), Harvard University (MH090948), Stanford University (MH101782), Washing-
424 ton University (MH101810), and to the University of Pennsylvania (MH101822). The datasets
425 used for the analyses described in this manuscript were obtained from dbGaP at [http://](http://www.ncbi.nlm.nih.gov/gap)
426 www.ncbi.nlm.nih.gov/gap through dbGaP accession number phs000424.v8.pht002741.v8.p2.

427 Author Contributions

428 J.K and A.Q.F designed the study, J.K developed the tools and analyzed the data, and J.K and
429 A.Q.F wrote the manuscript.

Bibliography

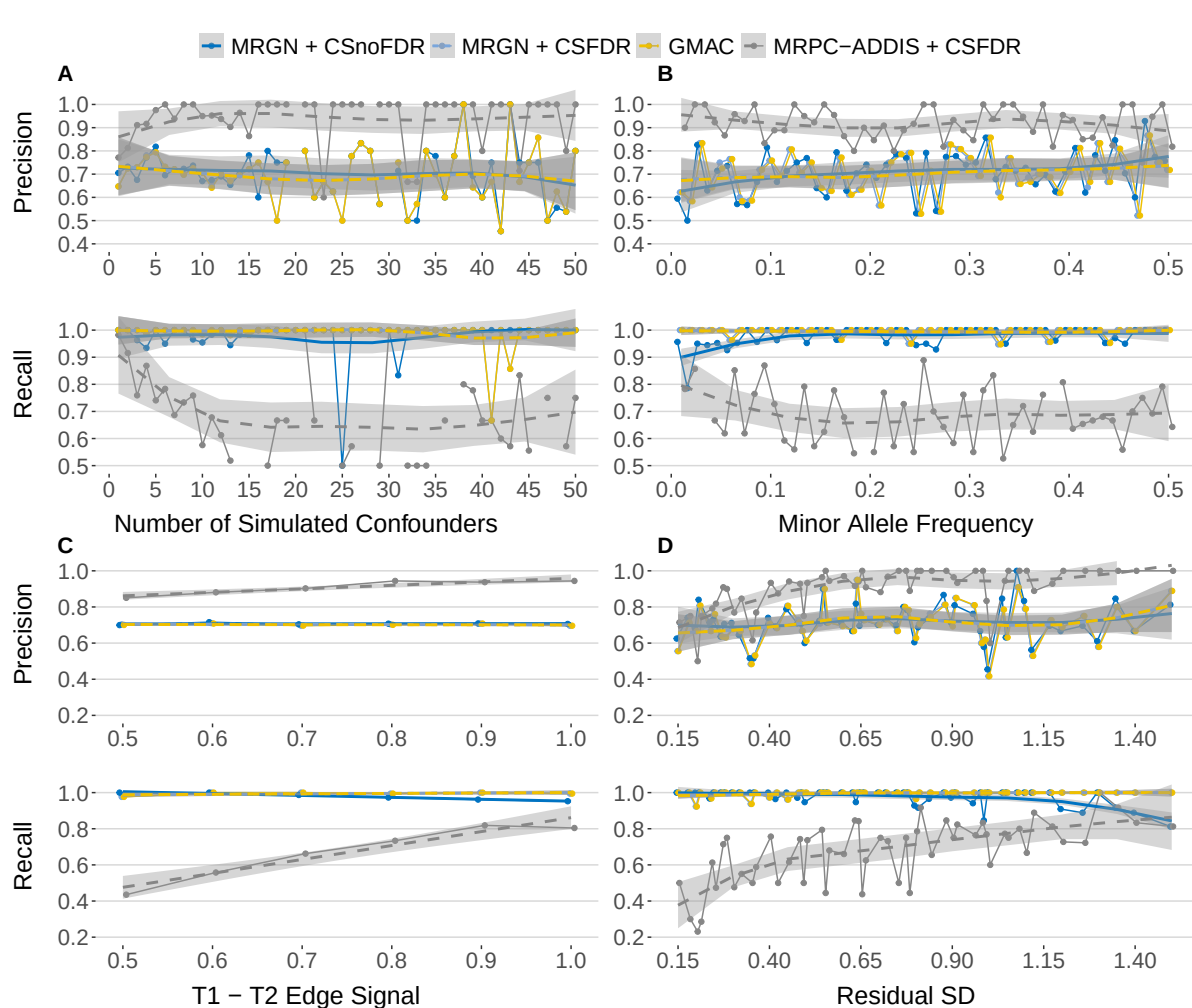
- [1] Diogo M Camacho, Katherine M Collins, Rani K Powers, James C Costello, and James J Collins. Next-generation machine learning for biological networks. Cell, 173(7):1581–1592, 2018.
- [2] Jean-Baptiste Pingault, Paul F O’reilly, Tabea Schoeler, George B Ploubidis, Frühling Rijdsdijk, and Frank Dudbridge. Using genetic data to strengthen causal inference in observational research. Nature Reviews Genetics, 19(9):566–580, 2018.
- [3] Chunming Xu and Scott A Jackson. Machine learning and complex biological data. Genome biology, 20(76):1–4, 2019.
- [4] Paola Lecca. Machine learning for causal inference in biological networks: Perspectives of this challenge. Frontiers in Bioinformatics, 1:746712, 2021.
- [5] George Davey Smith and Gibran Hemani. Mendelian randomization: genetic anchors for causal inference in epidemiological studies. Human Molecular Genetics, 23(R1):R89–R98, 2014.
- [6] Jason E Aten, Tova F Fuller, Aldons J Luskis, and Steve Horvath. Using genetic markers to orient the edges in quantitative trait networks: the neo software. BMC systems biology, 2(1):1–21, 2008.
- [7] Md Bahadur Badsha and Audrey Qiuyan Fu. Learning causal biological networks with the principle of mendelian randomization. Frontiers in genetics, 10:454043, 2019.
- [8] Azam Yazdani, Akram Yazdani, Ahmad Samiei, and Eric Boerwinkle. Generating a robust statistical causal structure over 13 cardiovascular disease risk factors using genomics data. Journal of Biomedical Informatics, 60:114–119, 2016.
- [9] Adriaan-Alexander Ludl and Tom Michoel. Comparison between instrumental variable and mediation-based methods for reconstructing causal gene networks in yeast. Molecular Omics, 17(2):241–251, 2021.
- [10] Tyler J VanderWeele, Eric J Tchetgen Tchetgen, Marilyn Cornelis, and Peter Kraft. Methodological challenges in Mendelian randomization. Epidemiology, 25(3):427–435, 2014.

- 458 [11] Connor A Emdin, Amit V Khera, and Sekar Kathiresan. Mendelian randomization.
459 Journal of American Medical Association, 318(19):1925–1926, 2017.
- 460 [12] Md Bahadur Badsha, Evan A Martin, and Audrey Qiuyan Fu. MRPC: An R package for
461 inference of causal graphs. Frontiers in Genetics, 12, 2021.
- 462 [13] Jarred Kvamme, Md Bahadur Badsha, Evan A Martin, Jiayu Wu, Xiaoyue Wang, and Au-
463 drey Qiuyan Fu. Causal network inference of cis- and trans-gene regulation of expression
464 quantitative trait loci across human tissues. Genetics, page iyaf064, 04 2025.
- 465 [14] Joshua Millstein, Bin Zhang, Jun Zhu, and Eric E Schadt. Disentangling molecular rela-
466 tionships with a causal inference test. BMC Genetics, 10(1):23, 2009.
- 467 [15] Huang Wang and Fred A van Eeuwijk. A new method to infer causal phenotype networks
468 using QTL and phenotypic information. PLoS One, 9(8):e103997, 2014.
- 469 [16] Fan Yang, Jiebiao Wang, Brandon L Pierce, Lin S Chen, François Aguet, Kristin G Ardlie,
470 Beryl B Cummings, Ellen T Gelfand, Gad Getz, Kane Hadley, et al. Identifying cis-
471 mediators for trans-eqtls across many human tissues using genomic mediation analysis.
472 Genome research, 27(11):1859–1871, 2017.
- 473 [17] Lingfei Wang and Tom Michoel. Efficient and accurate causal inference with hidden
474 confounders from genome-transcriptome variation data. PLoS computational biology,
475 13(8):e1005703, 2017.
- 476 [18] Lingfei Wang, Pieter Audenaert, and Tom Michoel. High-dimensional Bayesian network
477 inference from systems genetics data using genetic node ordering. Frontiers in genetics,
478 10:1196, 2019.
- 479 [19] Richard Howey, So-Youn Shin, Caroline Relton, George Davey Smith, and Heather J
480 Cordell. Bayesian network analysis incorporating genetic anchors complements conven-
481 tional Mendelian randomization approaches for exploratory analysis of causal relation-
482 ships in complex data. PLoS Genetics, 16(3):e1008198, 2020.
- 483 [20] Wonseok Oh, Junghyun Jung, and Jong Wha J Joo. MR-GGI: accurate inference of
484 gene–gene interactions using Mendelian randomization. BMC Bioinformatics, 25(1):192,
485 2024.
- 486 [21] Stephen Burgess, Rhian M Daniel, Adam S Butterworth, Simon G Thompson, and EPIC-
487 InterAct Consortium. Network mendelian randomization: using genetic variants as in-
488 strumental variables to investigate mediation in causal pathways. International journal of
489 epidemiology, 44(2):484–495, 2015.
- 490 [22] Nayang Shan, Zuoheng Wang, and Lin Hou. Identification of trans-eqtls using mediation
491 analysis with multiple mediators. BMC bioinformatics, 20(3):87–97, 2019.

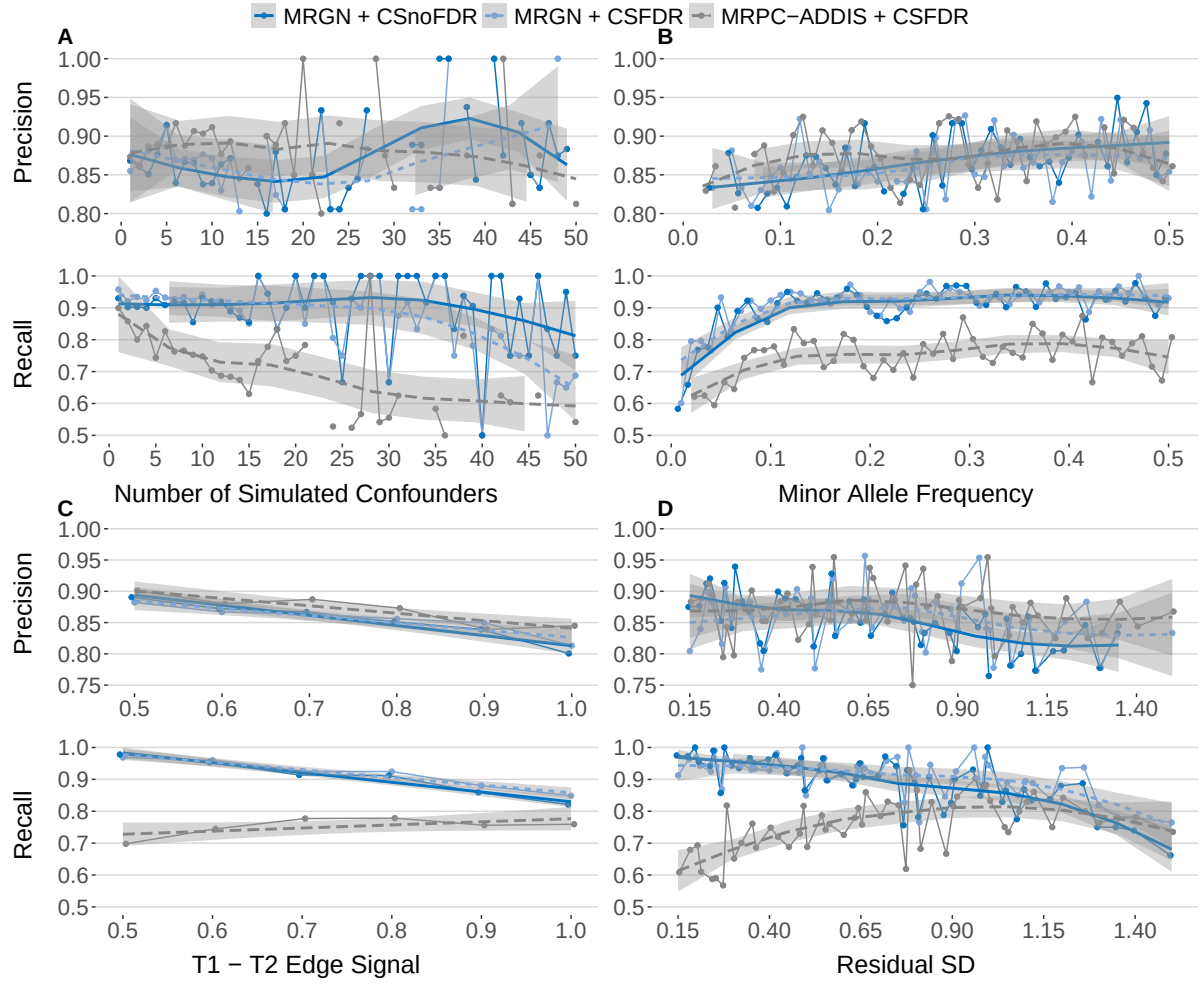
- 492 [23] David Amar, Nasa Sinnott-Armstrong, Euan A Ashley, and Manuel A Rivas. Graphi-
493 cal analysis for phenome-wide causal discovery in genotyped population-scale biobanks.
494 Nature communications, 12(1):350, 2021.
- 495 [24] Siming Zhao, Wesley Crouse, Sheng Qian, Kaixuan Luo, Matthew Stephens, and Xin He.
496 Adjusting for genetic confounders in transcriptome-wide association studies improves
497 discovery of risk genes of complex traits. Nature Genetics, 56(2):336–347, 2024.
- 498 [25] The GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across
499 human tissues. Science, 369(6509):1318–1330, 2020.
- 500 [26] John D Storey. The positive false discovery rate: a bayesian interpretation and the q-value.
501 The annals of statistics, 31(6):2013–2035, 2003.
- 502 [27] Andrej-Nikolai Spiess. propagate: Propagation of uncertainty using higherorder taylor
503 expansion and monte carlo simulation. R package: 1.0-6, 2018.
- 504 [28] Julie Josse, François Husson, et al. missMDA: a package for handling missing values in
505 multivariate data analysis. Journal of Statistical Software, 70(1):1–31, 2016.
- 506 [29] Md Bahadur Badsha and Audrey Qiuyan Fu. Learning causal biological networks
507 with the principle of Mendelian randomization. Frontiers in Genetics, 10:460, 2019.
508 PMC6536645.
- 509 [30] Peter Spirtes, Clark N Glymour, and Richard Scheines. Causation, prediction, and search.
510 MIT press, 2000.

511 **Tables**

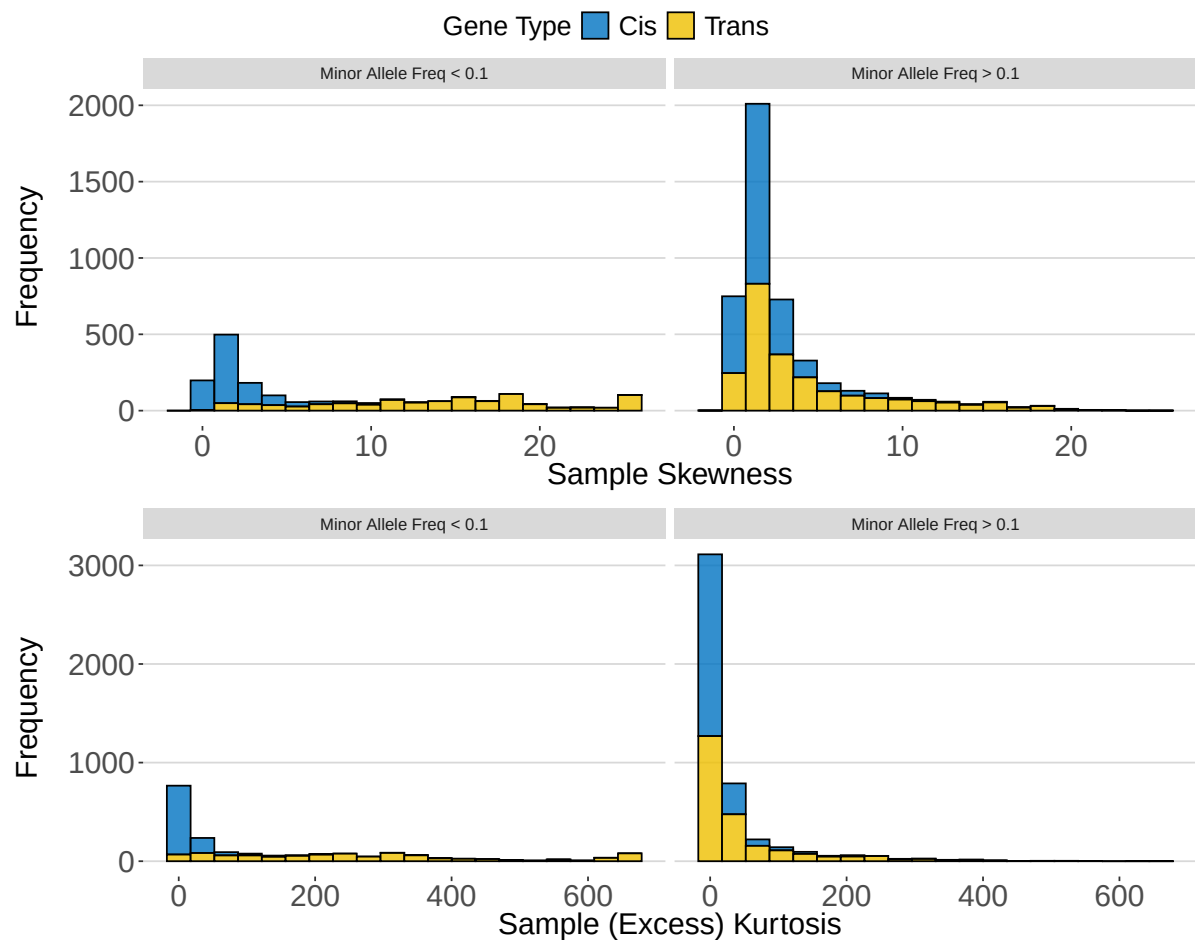
512 **Supplementary Material**



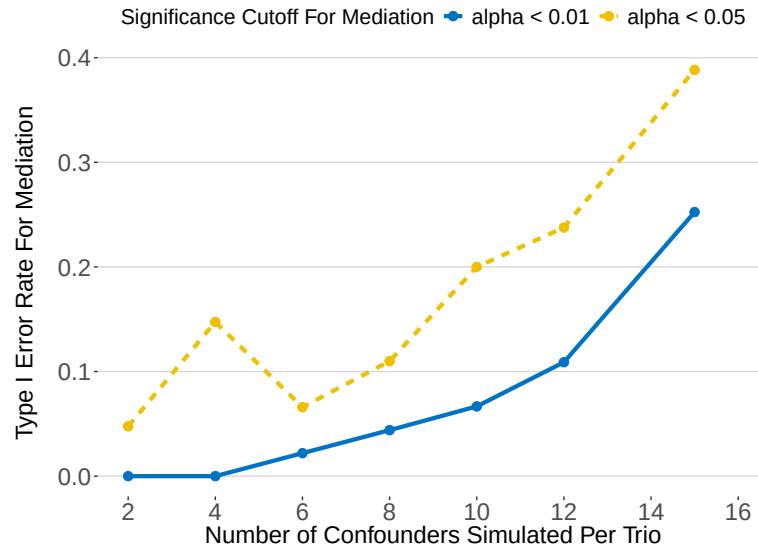
Supplementary Figure 1: Precision-recall curves for predicting the $T_1 - T_2$ edge across multiple simulation parameters. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



Supplementary Figure 2: Precision and recall curves of edge-based prediction performance for MRGN, and MRPC with confounder selection. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



Supplementary Figure 3: Skewness and kurtosis for cis and trans genes by allele frequency. Stacked histograms showing the sampling distributions of sample skewness and excess kurtosis for cis and trans genes in GTEx trios. Results are segregated by the minor allele frequency of the genetic variant, highlighting differences between common and rare alleles.



Supplementary Figure 4: Results of simulations investigating type I error rate (false mediation) under the null model. Trios were simulated under the null model for each of seven scenarios: 2, 4, 6, 8, 10, 12 or 15 confounders, excluding common child and intermediate variables. The GMAC method was applied to trios in each scenario using CS-q confounder selection.

514 Supplementary tables

Supplementary Table 1: A complete list of simulated trios, their simulation parameters, and inference results under MRGN, MRPC-ADDIS, and GMAC.

Supplementary Table 2: Different confounder selections were applied in MRGN and GMAC. The table summarizes the number of trios inferred to detect (or miss) the presence (or absence) the $T_1 - T_2$ edge, as well as recall, precision, and actual type I and II error rates.

Supplementary Table 3: All analyzed GTEx whole-blood trios, including genetic annotations (locus, variant ID, gene name, etc.) and inference results from MRGN, MRPC-ADDIS, and GMAC.

Supplementary Table 4: This table compares MRGN, MRPC, and GMAC on results from application to GTEx whole blood tissue. Results compared between MRGN with CS- α confounder selection to MRPC and GMAC.

Supplementary Text 1: Detailed supplementary text to explain the four simulation scenarios presented under the section “MRGN is robust to model misspecification in causal mediation”. This text also includes supplementary examples with output illustrating differences in causal inference between MRGN, GMAC, and MRPC on selected GTEx and simulated trios.